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AJR FOCUS ON:

Pleural Effusion: Characterization with CT Attenuation Values and CT Appearance

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OBJECTIVE. The purpose of this study was to assess the utility of CT in characterizing pleural effusions on the basis of attenuation values and CT appearance.

MATERIALS AND METHODS. We retrospectively analyzed 100 pleural effusions in patients who underwent chest CT and diagnostic thoracentesis within 48 hours of each other. On the basis of Light's criteria, effusions were classified as exudates or transudates using laboratory biochemistry markers. The mean value in Hounsfield units of an effusion was determined using a region of interest on the three slices with the greatest quantity of fluid. All CT scans also were reviewed for the presence of additional pleural features such as fluid loculation, pleural thickening, and pleural nodules.

RESULTS. Twenty-two of the 100 pleural effusions were transudates and 78 were exudates. The mean attenuation of the exudates (7.2 HU; [SD] 9.4 HU; range, 21–28 HU) was not significantly lower than the mean attenuation of the transudates (10.1 HU; 6.9 HU; range, 0.3–32 HU), ($p = 0.24$). None of the additional CT features accurately differentiated exudates from transudates ($p > 0.1$). Fluid loculation was found in 58% of exudates and in 36% of transudates. Pleural thickening was found in 59% of exudates and in 36% of transudates.

CONCLUSION. The clinical use of CT attenuation values to characterize pleural fluid is not accurate. Although fluid loculation, pleural thickness, and pleural nodules were more commonly found in patients with exudative effusions, the presence of these features does not accurately differentiate between exudates and transudates.

Pleural effusion is a common clinical finding with many potential causes [1]. The first step in the evaluation of a pleural effusion is to determine whether the pleural fluid is a transudate or an exudate. The formation of a transudate usually results from increased capillary hydrostatic pressure or from decreased colloid osmotic pressure. The main cause of transudates is usually congestive heart failure (CHF). The formation of an exudate usually results from an increased permeability of the microvascular circulation, generally due to inflammatory or neoplastic processes [2]. Diagnostic thoracentesis is routinely used as the first step in the characterization of a pleural effusion as transudate or exudate. Biochemical, cytologic, and microbiological analysis of the effusion can lead to determination of the cause in most cases [1].

Diagnostic thoracentesis carries small but significant risks. Major complication rates of thoracentesis range from 12% to 30% [3, 4]. The most common complication is pneumothorax, occurring in around 12% of the procedures with 5% of patients requiring chest tube insertion [4]. Minor complications include pain, shortness of breath, cough, and hematoma. Relative contraindications for thoracentesis include a bleeding diathesis, systemic anticoagulation, a small volume of pleural fluid, mechanical ventilation, inability of the patient to cooperate, and cutaneous disease such as herpes zoster or other infection at the needle insertion site [5].

A noninvasive method to characterize pleural fluid would be valuable for avoiding the potential risks associated with thoracentesis and may help to guide therapy. In addition, such a method could help with diagnosis in patients with a contraindication for the invasive procedure. CT is frequently used to assess patients with pleural abnormalities associated with neoplasm, pneumonia, and empyema. Exudates usually contain high levels of protein, lactate dehydrogenase (LDH), bilirubin, and cholesterol [1], thus

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potentially they may show greater attenuation values on CT.

In the past, the finding of pleural thickening at CT in patients with pneumonia or neoplasm was found to be highly indicative for the presence of an exudate [6]. Loculation, pleural nodules, and increased density of extrapleural fat were more frequently encountered in CT of patients suffering from empyema [7]. To our knowledge, all previous studies regarding CT features of pleural effusion were performed more than a decade ago, and the equipment used in these studies was not modern compared with current CT scanners. A previous study found that the use of CT attenuation values to characterize pleural fluid is moderately accurate [8]. The purpose of our study was to assess the utility of modern, MDCT examinations in characterizing pleural effusions on the basis of attenuation values and CT appearance.

Materials and Methods

Patients

The research protocol was approved by the local institutional review board and informed consent was waived. This retrospective study included patients who underwent thoracentesis between January 2006 and December 2007. Patients were selected for the study if they met the following criteria: presence of a pleural effusion on chest CT, pleural and serum LDH and total protein values obtained at thoracentesis, and CT and thoracentesis performed within 48 hours of each other.

Pleural effusions were classified based on Light's criteria [9], which categorize an effusion as an exudate if it shows one or more of the following features: pleural fluid total-protein-to-serum-total-protein ratio > 0.5, pleural fluid LDH-to-serum-LDH ratio > 0.6, and pleural fluid LDH greater than two thirds of the upper limit of normal LDH (620 IU/L at our institution). In addition to these markers, we determined the most likely cause of the pleural fluid, such as CHF, pneumonia, or malignancy, using clinical information from the patient's files combined with additional laboratory data such as pleural and serum Gram stains, cultures, and cytologic examination. Exudative pleural effusions associated with pneumonia were separated into two groups for further subgroup analysis based on the necessity of chest tube insertion. Complicated parapneumonic effusion was defined as the presence of empyema characterized by macroscopic pus or positive pleural fluid culture or when there were very high pleural LDH (> 1,000 IU), low pH (< 7.1), or low glucose (< 40 mg/dL) values [10]. Malignant pleural effusion was defined when positive pleural cytology or biopsy was found in patients with malignant disease.

Scanning Parameters

CT was performed on all patients using either of two 16-MDCT scanners (LightSpeed, GE Healthcare [n = 33] or Brilliance, Philips Healthcare [n = 28]) or a 64-MDCT scanner (Brilliance, Philips Healthcare [n = 39]). Standard scanning parameters of chest CT for each machine were used with slice thickness of 3.75, 5, 1, or 1.25; pitch of 1; 120 kV; and automated mAs.

IV contrast material was not administered when renal function tests were abnormal, in patients with high risk for contrast nephropathy (dehydration, diabetes mellitus, etc.), in patients with an allergy to contrast material, or when the indication for CT did not necessitate the use of contrast material. IV contrast material was administered in 53 patients. In 30 patients, standard chest examination was performed after a standard injection protocol (100 mL of iopamidol 300) with an injection rate of 2.5 mL/s, and in 23 patients, an angiographic examination was performed with 120 mL of IV contrast material (iopamidol 300) given at a rate of 4 mL/s. The image data were assessed on our PACS system (Centricity, GE Healthcare).

Data Acquisition

All CT scans were reviewed by two radiologists with 20 and 18 years of experience who were blinded to the clinical and laboratory information. Cases of interobserver disagreement were resolved by consensus. To evaluate the pleural fluid attenuation, we used the average measure of the three slices with the greatest amount of fluid, which was determined by the largest anteroposterior diameter of the effusion (mean for patients was 64 mm; range, 20–175 mm). A region of interest was placed for measurement of Hounsfield unit values of the maximal amount of fluid on each slice of the three slices used (Fig. 1). The radiologist took care not to include adjacent ribs, lung parenchyma, or areas of pleural thickening.

All CT scans were also reviewed for the presence of additional pleural effusion features. An effusion was considered loculated when it showed septations, was compartmentalized or accumulated in a fissure or a nondependent portion of the pleura, or when it showed a convex shape facing the lung parenchyma. Otherwise, a concave-shaped effusion that accumulated in the dependent portion of the pleural space was classified as free pleural fluid. The presence of pleural nodules in the parietal or visceral pleura was also considered. The CT scans were also evaluated for the presence of visceral or parietal pleural thickening. Parietal pleural thickening was diagnosed only if a pleural line was visible internal in relation to the ribs and thicker than 4 mm. Visceral pleural thickening was diagnosed only if a pleural line was visible on the surface of the lung adjacent to the fluid and could be reliably differentiated from the compressed lung. Extrapleural fatty tissue was evaluated for the presence of increased attenuation. It was divided into either the same density as that of the chest wall fat or increased density.

Statistical Analysis

Continuous data are described as mean value ± SD. The difference between the mean attenuation values of transudates and exudates was evaluated using a Mann-Whitney test. The association between the mean Hounsfield units of the pleural effusions and pleural total protein, pleural–serum total protein, pleural LDH, and pleural–serum LDH was examined individually using Pearson's correlation coefficient. A receiver operating characteristic (ROC) curve was constructed to determine the accuracy of attenuation values in the identification of exudates using the area under the ROC curve (A_z). The ROC curve was also used to determine the optimal threshold value (minimal false-negative and false-positive results) to classify transudates and exudates on the basis of mean

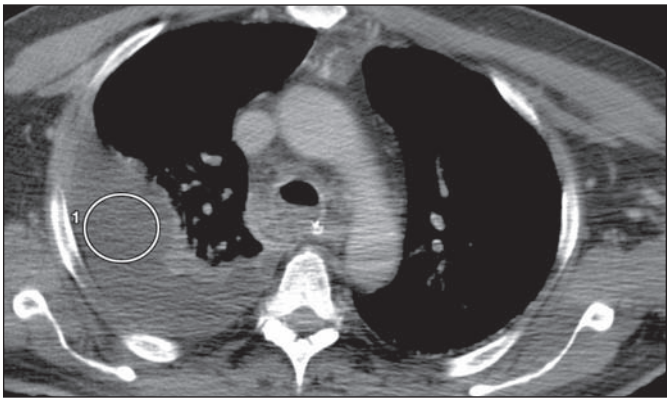


Fig. 1—Contrast-enhanced CT scan of chest at level of aortic arch in 73-year-old man with right-sided empyema shows loculated pleural effusion (circle) on right side 934 mm^2 in size with mean of –13.4 HU and SD of 27.55.

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Hounsfield units. Each additional pleural effusion CT feature was evaluated for the ability to predict exudates using a chi-square test. The usefulness of each feature for identifying exudates and transudates was also evaluated by calculating the sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV). Values for p of < 0.05 were considered significant.

Results

A total of 107 patients underwent chest CT and thoracentesis within 48 hours between January 2006 and December 2007. Seven patients had insufficient laboratory data to characterize their effusion according to Light's criteria and were excluded from the study. In the population of 100 patients, there were 62 men and 38 women (age range, 17–94 years; mean age, 62.4 years).

Attenuation

According to Light's criteria, 22 of the 100 pleural effusions were transudates and 78 were exudates. Table 1 summarizes the different causes of pleural effusions and their respective mean attenuation in Hounsfield units. The mean attenuation of exudates (7.2 HU; [SD] 9.4 HU) was not significantly lower than that of transudates (10.1 HU; 6.9 HU), (p = 0.24). The attenuation of exudates ranged from –21 to 28 HU and the attenuation of transudates ranged from 0.3 to 32 HU. All 13 negative attenuation values were found in exudates (Fig. 1). Eight of these patients were found to have acute pneumonia, three had chronic effusion due to malignant disease, and two had pleural effusion of unknown cause. There was a mild but significant positive relationship between mean Hounsfield units and pleural total protein (Table 2).

An ROC curve for the accuracy of attenuation values in the identification of exudates is shown in Figure 2. The overall accuracy for identifying exudates was low, (A_z = 0.582; 95% CI, 0.479–0.680), resulting primarily from the overlap in scores between 0 and 13 HU, which constituted 64% (64/100) of effusions. The optimal threshold value for exudates was determined to be ≤ 8.5, which showed a sensitivity of 55.1% (43/78) and specificity of 68.2% (15/22) (Fig. 3).

Additional CT Features of Pleural Effusion

The sensitivity, specificity, PPV, and NPV of CT findings for diagnosis of exudates versus transudates is elaborated in Table 3. We found no statistically significant correlation between any of the additional pleural effusion CT features and the ability to differentiate exudates from transudates (p > 0.1 for all). We could not evaluate the extrapleural fatty tissue for the presence of increased attenuation because most of the patients had only a minimal amount of extrapleural fat.

Loculation

Of the 22 transudates, eight showed a loculated pleural effusion (36%) (Fig. 4) compared with 45 of 78 exudates (58%). Twenty of the 21 complicated parapneumonic effusions (including empyemas) showed loculation (95%). Nine of the 19 malignant effusions showed loculation (47%).

Pleural Nodules

Pleural nodules were present in one of 22 transudates (5%) (Fig. 5) and in 10 of 78 exudates (13%). Six of the 19 malignant effusions had pleural nodules (32%) (Fig. 6). Four other patients with exudates and pleural nodules did not have malignant disease.

Pleural Thickening

Of the 22 transudates, eight showed pleural thickening (36%) (Fig. 5) compared with 46 of 78 exudates (59%). Sixteen of the 21 complicated parapneumonic effusions (including empyemas) showed pleural thickening (76%).

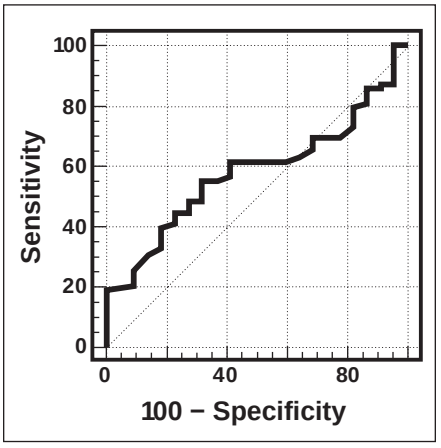


Fig. 2—Graph shows receiver operating characteristic (ROC) curve plotting 100 – specificity (x axis) against sensitivity (y axis). Overall accuracy was low, with area under ROC curve of 0.582 and standard error of 0.071.

TABLE 1: Summary of Causes and Attenuation Values for Pleural Effusions

Characteristic	No. of Lesions (n = 100)	Mean/SD (HU)
All effusions	100	7.9/9.0
Transudates	22	10.1/6.9
Congestive heart failure	20	11.1/6.4
Other[a]	2	0.3/0
Exudates	78	7.2/9.4
Malignancy	19	7.2/8.5
Parapneumonic effusion	19	5.8/9.9
Complicated parapneumonic effusion and empyema	21	8.1/10.2
Pulmonary embolism	3	5.0/5.5
Other[b]	13	10.8/8.0
Unknown	3	–3.3/12.4

[a]Other causes of transudates include nephrotic syndrome (n = 1) and hypothyroidism (n = 1).
[b]Other causes of exudates include hemothorax (n = 2), tuberculosis (n = 2), viral infection (n = 2), abdominal abscess (n = 2), congestive heart failure (n = 2), pancreatitis (n = 1), rheumatoid arthritis (n = 1), and Q fever (n = 1).

TABLE 2: Relationship Between Laboratory Markers and Attenuation

Laboratory Marker	Correlation (r)	p
Pleural total protein	0.22	0.03
Pleural–serum total protein	0.14	0.18
Pleural LDH	–0.06	0.52
Pleural–serum LDH	–0.05	0.65

Note—LDH = lactate dehydrogenase.

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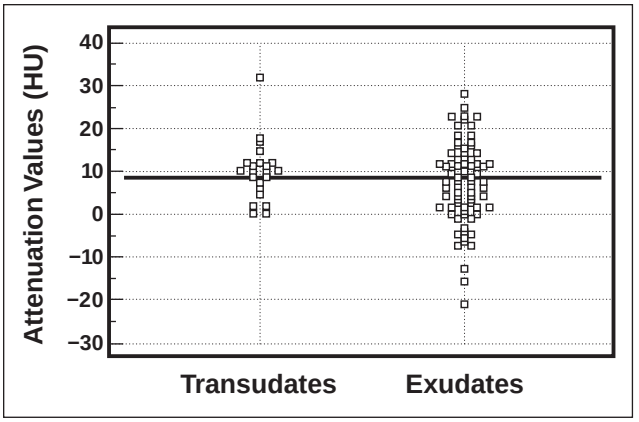


Fig. 3—Interactive dot diagram plotting attenuation values on *y* axis for exudates and transudates shows level of 8.5 HU was the optimal threshold value, with a sensitivity of 55.1% (43/78) and specificity of 68.2% (15/22).

Relation Between CT Protocol and CT Features

Ten of 22 patients with transudates (45%) and 43 of 78 patients with exudates (55%) received IV contrast material. The mean attenuation of the 43 patients with exudates who received contrast was 7.2 HU, and the mean attenuation of the 35 patients who did not receive contrast material was also 7.2 HU. The mean attenuation of the 10 patients with transudates who received contrast material was 6.4 HU compared with 13.2 HU in the 12 patients with transudates who did not receive contrast material. Loculation was found in 28 of 47 patients who did not receive contrast material (60%) compared with 25 of 53 patients who received contrast material (47%). Pleural nodules were found in four of 47 patients who did not receive contrast material (9%) compared with seven of 53 patients who received contrast material (13%). Pleural thickening was found in 24 of 47 patients who did not receive contrast material (51%) compared with 30 of 53 patients who received contrast material (57%). Thirty-one patients underwent thoracentesis before CT and 69 underwent CT before thoracentesis (both within 48 hours). The mean attenuation of patients in whom thoracentesis preceded the CT was mildly lower (7.7 HU compared with 7.9 HU).

Discussion

The pleural space normally contains between 7 and 14 mL of fluid [10]. An effusion will accumulate in the pleural space whenever the rate of fluid formation exceeds the rate of fluid removal. This can occur either because of an elevated net hydrostatic pressure gradient (transudation) or because of an increased permeability of the pleural vessels (exudation) [11]. Pleural effusions also may be caused by delayed resorption (lymphadenopathy, radiation therapy). Diagnostic thoracentesis is performed to determine the specific cause of a pleural effusion and supplies biochemistry measurements (e.g., protein and LDH) that help to separate effusions into transudates and exudates. Further analysis such as cytology and cultures may help in establishing the specific cause of the effusion [1]. Although thoracentesis is considered a relatively safe procedure, it is associated with risks such as pneumothorax and has several contraindications such as coagulopathy [3–5]. Finding an efficient noninvasive technique to help characterize pleural effusions is challenging and would be beneficial for many patients. Nevertheless, our study found that pleural fluid attenuation on chest CT had no significant role in characterizing pleural effusions as exudates or transudates or in helping to differentiate a complicated parapneumonic effusion necessitating chest tube insertion from a regular parapneumonic effusion. The mean CT attenuation values in our study were almost identical for both types of effusion. We found a considerable overlap in values with the majority of effusions in the 0–13 HU range (64%). Other additional CT features of pleural effusion, such as loculation, pleural nodules, and pleural thickening, did not accurately predict the presence of transudates or exudates in our study.

Before this study, we expected to see increased attenuation in exudates because exudative fluid usually contains high levels of protein, LDH, and bilirubin, which all potentially can show increased attenuation on a CT scan. Only one clinical study has been published on the characterization of pleural effusions using CT attenuation. Nandalur et al. [8] examined 145 patients and found that the mean attenuation of exudates

TABLE 3: Sensitivity, Specificity, Positive Predictive Value (PPV), and Negative Predictive Value (NPV) of CT Findings for Diagnosis of Exudative (*n* = 78) Versus Transudative (*n* = 22) Pleural Effusions

CT Finding	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)
Loculation	58	64	85	30
Pleural nodules	13	95	91	24
Pleural thickening	59	64	85	30

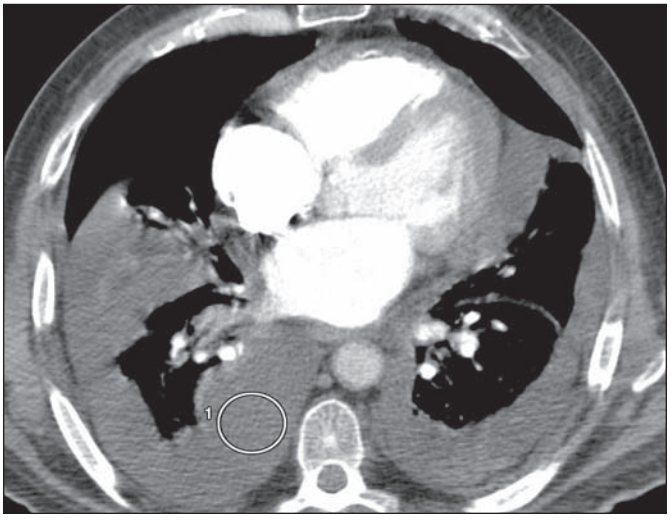


Fig. 4—Contrast-enhanced axial CT scan of thorax in 80-year-old woman with congestive heart failure showing bilateral loculated pleural effusion. CT density measurement of right effusion (*circle*) 691 mm^2 in size was 14.5 HU with SD of 19.8. Fluid was proven to be transudate at thoracentesis.

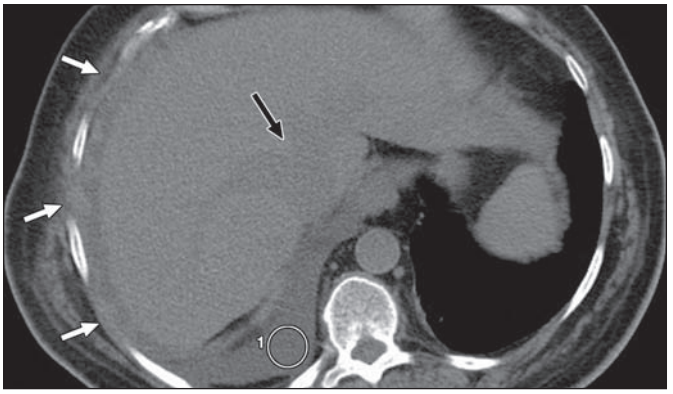


Fig. 5—Unenhanced CT scan of chest in 69-year-old woman with congestive heart failure shows right-sided pleural effusion with marked, nodular pleural thickening (*white arrows*). CT density of 330-mm^2 effusion (*circle*) was 11.3 HU with SD of 13.7, and thoracentesis revealed transudate. Note dilatation of inferior vena cava and hepatic veins consistent with right-sided heart failure (*black arrow*).

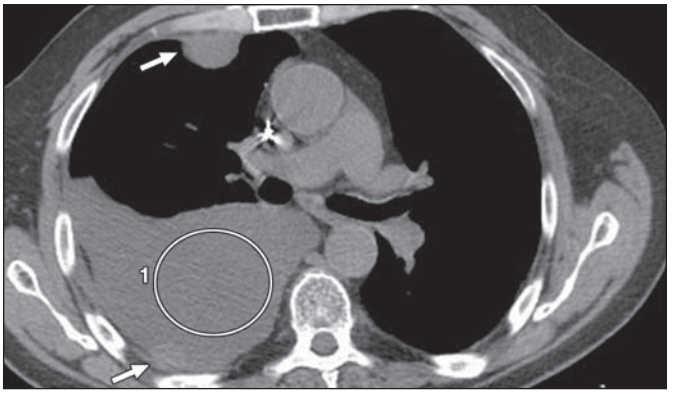


Fig. 6—Unenhanced CT scan of thorax in 79-year-old man shows relatively low density (8.8 HU; SD, 13.3) 2,549-mm^2 right-sided pleural effusion (*circle*). Patient had pleural exudate secondary to colon carcinoma with pleural metastasis (*white arrows*).

was 17.1 HU compared with a mean attenuation of 12.5 HU for transudates. A modest but significant positive relationship between mean Hounsfield units and pleural protein and LDH was also found. The authors concluded that the overall accuracy of CT attenuation was moderate, with an optimal threshold value of 13.4 HU that showed a specificity of 71% and sensitivity of 83% for differentiating transudates from exudates.

In our study, mean attenuation of exudates was 7.2 HU compared with 10.1 HU for transudates, and the overall accuracy for identifying exudates was low (A_z = 0.582). Thus, CT attenuation value was found to be a poor indicator for characterizing an effusion. To increase the potential correlation between the fluid biochemistry findings and the CT attenuation value, we included in our study only patients who had undergone thoracentesis and CT within 48 hours as opposed to 7 days as in the work of Nandalur et al. [8]. Recent data show that pleural fluid obtained a few days after diuresis from patients with CHF can be misinterpreted as exudative when using Light's criteria [12] but not when the samples are obtained within 48 hours from the initiation of diuresis [13]. The effect of treatment in the time interval between thoracentesis and CT may also decrease the correlation between biochemistry markers and attenuation values.

None of the 145 effusions in the study of Nandalur et al. [8] showed a negative attenuation value. Our study found 13 effusions with negative attenuation, which were all found to be exudates. Previous studies have shown a significantly larger amount of cholesterol in exudates compared with transudates [14]. Hamm et al. [14] found that the elevated pleural cholesterol in exudates is the result of the underlying disease (e.g., malignancy, pneumonia, or tuberculosis) rather than a reflection of the serum cholesterol level. A possible explanation for the elevated pleural cholesterol may be greater cellular degeneration or increased pleural permeability in exudates compared with transudates [14]. Elevated cholesterol levels can also result from chylothorax, mainly due to trauma or lymphoma, and from pseudochylothorax, mainly due to tuberculosis, rheumatoid arthritis, or empyema [15]. Moreover, fat tissue is known to give negative attenuation values. Thus, the high concentration of protein in exudates that is expected to raise the attenuation values may be contradicted by the high cholesterol level that decreases the attenuation.

We found a small but significant positive relationship between mean Hounsfield units and pleural total protein (r = 0.22, p = 0.03), but we could not determine whether our negative attenuation fluids had high levels of cholesterol because pleural cholesterol was measured in only three of the 100 patients. Although both the study of Nandalur et al. [8] and our study were conducted using a similar methodology, we could not find a definitive explanation for the different attenuation values between the two studies. One possible explanation is the greater time gap between thoracentesis and CT in the former study, which may have enhanced the effect of treatment on pleural fluid biochemical markers and attenuation. Another possible explanation is the use of different CT scanners and protocols in the two studies.

Few studies have evaluated the efficacy of several pleural effusion CT features in differentiating between exudates and transudates. Arenas-Jiménez et al. [7] examined 211 patients and found that the presence of pleural thickening, pleural nodules, and loculation was highly specific for exudates. Of their 211 patients, 75 had pleural thickening, and all of these were exudates (42% sensitivity, 100% specificity). Aquino et al. [6] examined 86 patients with pleural effusion and found 37 cases of pleural thickening, with only one of these in a patient with transudates (61% sensitivity, 96% specificity). Waite et al. [16] found pleural thickening in 27 of 65 patients with exudates and none among 20 patients with transudates (42% sensitivity, 100% specificity). Wolek et al. [17] studied a series of 55 patients and found pleural thickening to be 50% specific and 100% sensitive for the existence of exudates. From all four of the studies mentioned, only one patient with a transudate was found to have pleural thickening. The suggested explanation for this finding was that this patient's history included a previous empyema [6]. Our study, on the other hand, found pleural thickening in eight of the 22 transudates (36%) compared with 46 of 78 exudates (59%).

Loculation of the effusion was found in 24 of 211 patients in the study of Arenas-Jiménez et al. [7], all of them in patients with exudates. Our study, on the other hand, found loculated pleural effusion in eight of the 22 transudates (36%) compared with 45 of 78 exudates (58%).

Both pleural thickening and loculation were found in more than one-third of the patients with transudates in our study. This clearly contradicts the finding of the four

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relatively large studies mentioned [6, 7, 16, 17]. A possible explanation for this difference may be that all four studies were performed 10 years ago or more. The quality and resolution of the CT images in our study were probably higher than those of the previous studies, thus elevating the sensitivity of these finding but decreasing their specificity in the present study. Furthermore, pleural thickening may be old or chronic and not related to the effusion studied on the patient's present admission. We believe that our results suggest that the clinical use of both loculation and pleural thickening for a definite differentiation between exudates and transudates should be discouraged.

Pleural nodules were found in 17 patients in the study of Arenas-Jiménez et al. [7], all of them in patients with exudates. In our study, pleural nodules were present in one of 22 transudates (5%) and in 10 of 78 exudates (13%). Although the presence of a pleural nodule was found to be highly specific, especially when the effusion is caused by malignancy, the low sensitivity of this finding limits its clinical usage.

Our study has several limitations. First, it is a retrospective study, and the thoracentesis and CT were not performed at the same time in most of our patients. As mentioned, diuresis can alter pleural biochemistries [12, 13]. Thus, some pleural fluids of patients with heart failure might have been misclassified as exudates. Moreover, treatment success or failure in patients with pneumonia might also influence biochemistries of pleural fluid or CT appearance. To minimize the effect of this limitation on our results, we limited the time interval between thoracentesis and CT to 48 hours. All of the previous clinical series mentioned had a maximal interval between CT and thoracentesis of from 7 to 20 days [6–8, 12, 13]. Another limitation is that chest CT in our study was performed using two different scanning parameters and three different scanners. In addition, some patients received IV contrast material and others did not. Nevertheless, there were no noticeable differences in measurements by the two radiologists, and the analysis presented in the results suggested that IV contrast material did not affect the attenuation values. Nonetheless, further studies may use a standard technique to improve the accuracy of the results. Finally, our study has a selection bias for effusions of malignant cause and effusions resulting from pneumonia, especially complicated parapneumonic effusions, because chest CT is generally not performed in conjunction with thoracentesis in patients suffering from CHF exacerbation. As a result, there was an overrepresentation of exudates in our study. Moreover, it is possible that the visualization of loculation or pleural thickening in a CT scan led to thoracentesis and thus affected the results.

It should be emphasized that CT is more sensitive than both conventional chest radiography and sonography for differentiating pleural fluid from pleural thickening, assessing fluid loculation, identifying focal masses, and assessing lung infiltrates [18]. The results of the present study should not discourage physicians from using this important tool for the management of patients with pleural effusion. CT can help with the diagnosis of the specific fluid cause and is a useful tool for guiding the exact placement of chest tubes when needed. Nevertheless, CT should not replace diagnostic thoracentesis when the latter is indicated.

In conclusion, CT attenuation values did not show any potential clinical value in the characterization of pleural fluid. Additional pleural CT appearance features such as fluid loculation, pleural thickness, and pleural nodules are not reliable in differentiating exudates from transudates, although their prevalence is higher among exudative effusions.

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